Bhumi Lipi

A trust layer that links deed registration to mutation, so land papers can be proven, not just shown

LandTech Innovation Hackathon 2026 · Round 1 Concept Note · Track: Digital Land Records, Verification & Digitization

Problem statement	**P2 - Deed & Ownership Verification (main)**. Built together with P3 (fraud and double-selling), P4 (registration-to-mutation workflow) and P5 (citizen pre-purchase check) as one lifecycle.
Idea in one line	Every sale becomes a signed link in a per-plot chain; the plot is locked until the land office approves mutation; anyone can verify a deed in seconds. It sits beside the existing registry and e-Mutation systems and does not replace them.
Prototype status	Working on 100% synthetic data: 53 automatic tests, browser walkthrough, independent verifier script, attack lab. Features not yet built are marked *Planned* throughout.

1. The problem in Bangladesh: how big, and where it breaks

Measure	Figure (source)
Court backlog	45,16,603 pending cases at 31 Dec 2024 [1]. Estimates of the land-related share range from about 60% to 80% (the Land Secretary has said almost 80%) [1][2]. No audited split exists.
Households affected	About 1 in 7 households was involved in a land dispute in a 2014 survey of 1,050 households in 14 districts; resolved cases took about 3 years, average pending time about 8 years [3]. Old, but the only such household study found.
Corruption in land services	TIB household survey 2023: 51% of households using land services met corruption; 77.4% of those seeking mutation paid bribes; average payment Tk 12,719 for mutation and Tk 19,037 for registration; Upazila Land Offices worst [4]. TIB 2025: average land-service bribe Tk 11,311 per household [5].
Volume of work	497 sub-registry offices; 36.72 lakh deeds registered in FY2017-18 (most land-related) [6]; about 22 lakh mutation applications a year per the Ministry of Land app [7]. Both are the scale a national system must handle.

Root cause: One land transaction touches three offices that keep three separate records. Deeds are registered by the Registration Directorate (Law Ministry), owner names in the government record are changed by AC Land offices (Land Ministry), and survey records are kept by the land records and survey department. Fraud lives in the gaps between them. Practitioner guides in Bangladesh describe the same patterns again and again [8]:

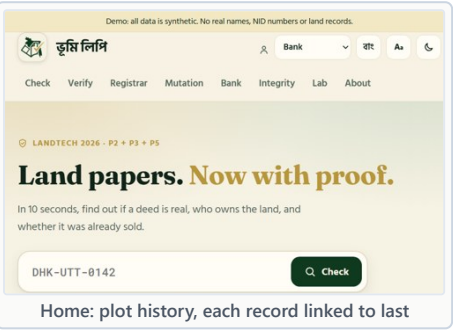
Where it breaks	How it happens	Effect
Forged or fake deed	A fabricated old (bia) deed that never existed in the registry volume book looks real on paper. A copy can be altered after issue.	Buyer, bank or court cannot tell in minutes; found years later in court.
Double sale	The same plot is sold again in the days or weeks between registration and mutation, before records catch up. Skipped mutation keeps old owner on revenue record.	Two buyers, one plot; a civil and often a criminal case.
Mutation delay	A separate application at a different office, with 3-4 visits before e-Mutation (now about 28 days in best case) [9] and widespread informal payments [4].	Owner stays exposed; delay is invisible and nobody is accountable.
Unreadable records	Four survey generations (CS, SA, RS, BS), ana-gonda units, broker-dependent checking.	Elders and first-time buyers are the most exposed.

2. What already exists, and the gap Bhumi Lipi fills

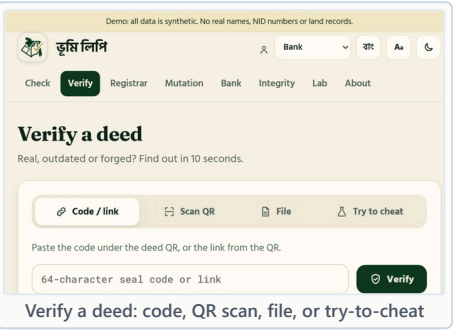
Bangladesh is not starting from zero. e-Mutation, e-Porcha, the digital record room, the Land Management Automation project and Bangladesh Digital Survey are moving records online [9][10][11]. From January 2026 the Land Secretary announced online seller confirmation for deed registration and mandatory buyer mobile numbers [12]. These reforms improve service delivery and identity. What none of them gives is: **(1) a way for an outsider to prove that a deed copy is exactly what the registry sealed; (2) a shared lock that blocks a second sale while mutation is pending; (3) a history that exposes any later edit, even by an insider.** Bhumi Lipi is designed as that missing trust layer, receiving a read-only feed from existing systems first, so no office has to give up its current process.

3. The solution (linked lifecycle)

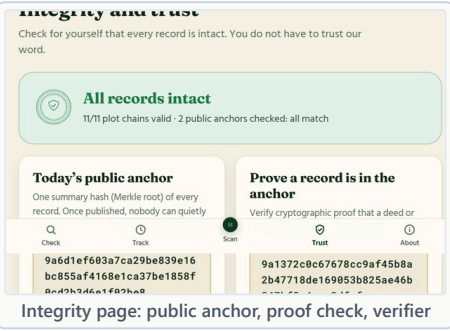
Step	What happens	What it prevents
Register	Sub-registrar enters the sale. 11 explainable rules run (fake seller, government/khas land, lien, rapid transfer, survey mismatch, ...).	Fake sellers and blocked plots stop before sealing.
Seal	Record is hashed (SHA-256), signed with the office key (Ed25519) and linked to the plot's previous record. QR printed on the deed.	Silent edits to history; forged copies.
Lock + mutation	Plot is locked; mutation case starts automatically with 8 checks and a tracking ID. Waiting-days chip turns amber at 14 days and red at 30.	Second sale in the gap; invisible delay.
Approve	AC Land officer decides from a queue; plot unlocks and the new owner is recorded in the same chain.	Two offices disagreeing about who owns the plot.
Verify	Anyone scans the QR or uploads the file: Genuine, Outdated (real, but a newer record exists) or Forged. Citizens get a Green/Amber/Red report, Simple mode for elders (3-word verdicts, big text, read-aloud, ana-gonda helper).	Broker dependence; buying on a fake paper.
Prove	A daily public Merkle root, an in-browser proof check, an independent verifier script, and an Attack Lab that attacks a private copy 5 ways, including an insider with a key.	"Trust us": anyone can check without our server.



Home: plot history, each record linked to last



Verify a deed: code, QR scan, file, or try-to-cheat



Integrity page: public anchor, proof check, verifier

4. How we sharpened the idea after studying the registry

Reading how registration and mutation really work exposed weak points in a purely technical design. Each is fixed below. Status is stated honestly.

Weak point found	Fix in the design	Status
A deed is not a title. Registration records documents, and a genuine deed can still come from someone who never owned the plot.	Every report separates two facts: deed integrity (Genuine/Outdated/Forged) and record agreement (does the deed's seller match the last recorded owner and survey record?). Green means "records agree and no block found", never "you own this".	Wording and mismatch flag built; full two-part report *Planned*
Garbage in. A seal cannot stop an impersonator or fake power of attorney registered through the front door.	Store a hash of the identity check (NID/biometric done at the registry), force officer review for power-of-attorney sales, and text the recorded owner when a sale or mutation starts on their plot, with a short objection window. Maker-checker: high-risk deals need a second officer's signature.	*Planned* (identity token, POA rule); owner alert and second signature *Planned*
Insider with a key can re-sign history.	Daily public anchor makes any rewrite visible (built). Add hardware keys per office, key revocation, three-office co-signing of each anchor, and anchor copies held by independent bodies (for example a court and a university).	Anchor and attack lab built; hardware keys and co-signing *Planned*
Mutation delay and discretion. Going online did not remove informal payments [4].	Make the clock visible and political: SLA set to the official service standard (sources cite 28-45 days, confirm with a land office), automatic escalation to UNO then ADC (Revenue) then DC, and a public per-office delay board. Software escalates; it never auto-approves.	Aging chip built; escalation ladder and board *Planned*
Old records cannot be re-proven by software.	Do not try to re-verify 100 years. Take a dated baseline seal of scanned balam (volume book) pages and khatians after officer review; from that day any change is detectable. Bangla OCR only with human review; synthetic scans in demos.	*Planned (Phase B)*
Parallel system risk. Offices will not run two systems.	Day-1 mode is a read-only adapter fed from registration and e-Mutation data; later two-way API. Open verifier and open format so no vendor lock-in.	*Planned*
Villages, privacy, law.	Offline-verifiable QR, union digital center help, SMS check. Public pages show only fields already public in khatians and mask the rest; the anchor holds hashes only. Legal weight of e-seals must be confirmed by land-law experts.	*Planned*; legal review needed

5. Architecture and tech stack

Layer	Technology and design choice
Browser	Plain JavaScript (no build step), Bangla and English, Simple mode, mobile tab bar, strict content security policy, textContent only.
Server / services	Python standard library HTTP layer; framework-free service layer (register, check, verify, decide, liens, integrity, anchor, lab). One third-party dependency: cryptography.
Crypto and rules	SHA-256, Ed25519 (one key per office), Merkle tree. 11 rules as pure functions with Bangla and English reasons, so every block or warning is explainable.
Database	SQLite now, PostgreSQL planned. Append-only triggers on records, anchors, mutation events and audit log; chain-guard trigger. Sealed block format frozen; changes additive only; feature flags for unfinished work.
Delivery	Dockerfile, render.yaml, CI, README, spec, threat model, demo script.

Scale check: 36.72 lakh deeds a year is about 10,000 a day, roughly one every 9 seconds on average. Hashing and signing at that rate is trivial for one ordinary server; the real storage cost is scanned documents, not the chain (see annex).

6. How much of the problem does it solve?

No honest percentage can be given yet. We found no public statistic that isolates forged-deed or double-sale cases from other land disputes, so any figure would be invented. What can be said is which causes the design reaches, and how a pilot would measure it:

Cause of land trouble	Reach	Why
Forged or altered deeds	Directly	For every deed sealed after adoption, a copy is checked against the signed record in seconds. Older deeds are covered only after baseline sealing.
Same plot sold twice	Directly	Lock between registration and mutation, shared chain, lien and khas blocks. Only works for plots and offices on the chain.
Buyer cannot check; broker dependence	Directly	Public check page, QR, Simple mode, printable report.
Registration-mutation mismatch	Mostly	One lifecycle, auto-start, mismatch flag. Depends on integration with e-Mutation.
Mutation delay and informal payments	Partly	Visible clocks, fewer hand-offs, escalation. Discretion and pay-to-speed-up need institutional enforcement too.
Survey errors (RS/BS), area mismatch	Partly	Mismatch warning now; lineage graph Planned. Correcting records still needs survey tribunals.
Inheritance disputes (warish)	Not yet	Succession workflow only Planned; family disputes need legal process.
Illegal possession, encroachment, force	No	Physical and enforcement problems; software only supplies evidence.
Court backlog	Indirect	Cleaner evidence may shorten some cases; no effect claimed.

Pilot success measures (baseline taken from pilot offices' last 12 months): double-sale attempts blocked; forged-copy detections; median days from registration to mutation start and to decision; share of cases over the official SLA; verification checks per month; officer and citizen feedback.

7. Plan for countrywide implementation

Stage	Scope	Work and exit test
1. Pilot	1-2 sub-registry offices with their AC Land offices (read-only feed)	Hardware keys, training, parallel run with paper process, baseline seal of one record room. Exit: zero chain breaks, officers can explain every block, law review done.
2. District	All offices in one district; real login; PostgreSQL	Owner SMS alert, escalation ladder, delay board, offline QR, union digital center help. Exit: SLA breaches measurably visible and falling.
3. Division	Several districts; two-way API	Connect to e-Mutation and registry systems by official agreement; ID integration; three-office anchor co-signing.
4. National	497 offices, one key each plus backups	Shared public anchor, open verifier, scheduled digitization of old khatians and deeds with OCR plus human review.

Real costs: hosting, database, hardware keys, officer training, scanning old records, and legal and security review. Open-source software and no per-transaction fee; we make no cost-saving claim because none has been measured. Needs from government: a data-sharing agreement, legal recognition of the e-seal, and a named owner in both ministries.

8. Limits, data rules and AI disclosure

- **The system is tamper-evident, not tamper-proof.** A wrongly approved fake that enters before sealing is not caught by software; identity checks and officer review reduce this risk but do not remove it. Not tested at national scale. Demo roles are not real login.
- **All data is synthetic.** No real names, NID or phone numbers, addresses or land records; nothing scraped from government portals; no live government system tested. AI coding tools were used and are listed with their role in the README; both students can explain the code.

9. Team and mentor details

Role	Name	DIU ID / Organization	Contact
Student 1 (Team Leader)	[fill in]	[fill in]	[fill in]
Student 2	[fill in]	[fill in]	[fill in]
Faculty mentor	[fill in]	[fill in]	[fill in]
Industry mentor	[fill in]	[fill in]	[fill in]

Team name: [fill in] | **Repository:** [link] | **Live link:** [link]

Sources. [1] Dhaka Tribune op-ed "Whose land is it anyway?" (court data to 31 Dec 2024). [2] Daily Star / scholarly estimates of land share (60-80%). [3] Daily Star, "One in 7 households facing land disputes" (PRI/BRAC, 2014). [4] TIB National Household Survey 2023, via The Financial Express. [5] TIB NHS 2025 executive summary. [6] Daily Star on TIB sub-registrar report (2019). [7] Ministry of Land e-Namjari app listing. [8] Practitioner guides (law-firm articles; not official statistics). [9] UN DESA e-Mutation case. [10] Land Management Automation / DLMS literature. [11] The Business Standard on Bangladesh Digital Survey. [12] Gonobarta, 24 Nov 2025, Land Secretary on 2026 registration changes.